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Ask of Koen

Proposal: Multi-Engine Classroom

Status: Proposal draft for discussion — supersedes [multi-engine-classroom-scenarios.md](#) (2026-09-28).

Revision: 2026-09-30f — Koen feedback: remove redundant edge prose and coaching-click table; coaching actions remain inside `kolibri_manage`.

Author: Steve (Lead Bot), 2026-09-29 (rev. 2026-09-30f)

Audience: Koen / IDEA leads

Companion capacity issue: [idea#159](#) — measure safe concurrent Kolibri video streams per instance

Why this exists

Koen asked for a clean framing of **why a school runs more than one Engine**, how that looks with a small named fleet, what classroom scenarios matter, and how a Markov usage graph can drive realistic tests later.

This document is a **proposal draft**. Implementation, discovery protocol, auth, Playwright harnesses, fleet claim scripts, and Engine code changes are explicitly out of scope here (see final section). The Markov usage graph itself is a **concrete state/action model** for later tests — not labeled conceptual.

Vision cite (this is already documented)

Prior drafting treated multi-Engine load distribution as an undocumented assumption. That was wrong.

It is already in the IDEA vision / solution description:

All Appdockers on the same LAN will automatically find one another and when a new Appdoker device is added, the catalog of available apps is automatically extended with the apps docked onto the new device.

Performance is optimized by adding Appdockers and redistributing the apps over the Appdockers.

Source: [agent-engine-dev/proposals/solution-description.md](#)

Referenced from `idea/docs/grok-bot-setup.md` as the Engine vision and intent doc.

The multi-Engine classroom story below **extends** that vision into classroom roles and usage scenarios; it does not invent the load-spread principle.

Why multi-engine (Koen's three reasons)

1. Distribute apps / load across Engines

The physical act is simple: **add an Engine** and **plug a demanding app into it**.

- Engines **autofind and connect** on the school LAN (vision: Appdockers find one another).
- The **integrated app list** looks the same whether the school has 1 Engine or N Engines.
- Users **never need to know** which Engine serves which app, or which Console they opened — any Console on any Engine can see the full picture.

This is the same physical metaphor as the rest of IDEA: capacity is improved by adding hardware and redistributing disks, not by asking teachers to become sysadmins.

2. Scale one app across Engines

Some apps hit a **per-instance / per-Pi limit**. Example: concurrent Kolibri video streams.

When one Kolibri instance cannot safely serve the whole class:

- Start a **second Kolibri instance** on another Engine.
- Assign students** across instances (class A → instance 1, class B or half the cohort → instance 2).
- Same pattern later for Nextcloud video / heavy media.

Capacity baseline for Kolibri ([idea#159](#), idea01 Pi 5 / 4 GB). Multi-quality **planning** streams per instance (LAN measured SAFE ≥128 for all four; planning leaves Wi-Fi / browser / Pi-4 headroom):

Quality	Resolution	Planning streams / instance
low	360p (640×360)	64
medium	480p (854×480)	32
prior / typical lesson	720p (1280×720)	32
high	1080p (1920×1080)	12

Default classroom planning figure remains **32 @ ~720p**; use 64 / 12 when content is known low / high. Load-test tooling: [agent-app-dev#9](#).

3. Per-class Engine with own instances — Console manages everything from anywhere

Each class (or teaching block) can have **its own Engine** hosting **class-only** app instances — e.g. Class 5A Nextcloud, Class 5A Kolibri content — without isolating operators.

Any Console on any Engine can manage **all apps across all Engines** simply. Teachers and operators are not trapped on “their” Pi’s Console; the mesh presents one school.

Initial 3-engine setup (concrete sketch)

Design target first: a **named three-Engine school**, then generalise to N.

Name	Role (story)	Typical disks / instances
idea-A	Shared / school golden hub	Shared apps everyone uses lightly: offline Wikipedia (Kiwix), school-wide Nextcloud “School Files”, maybe a small shared Kolibri channel library. Stable reference Engine.
idea-B	Class Engine — Grade 5A	Class-only Kolibri (Grade 5A lessons/quizzes), class Nextcloud (Group “Grade 5A”: view-only shares, File Drop, collaborative docs).
idea-C	Class Engine — Form 3	Class-only Kolibri (Form 3), class Nextcloud (Group “Form 3”). When Grade 5A video load overflows idea-B, a second Kolibri instance for 5A can also land here (reason 2).

Classroom story: Monde Primary (or any Marco site) starts with one Engine in the room. When video buffering and “limit to one group at a time” become the norm (field troubleshooting already warns about many devices), the school adds **idea-B** and **idea-C** as classroom Engines. Teachers still open any Console; students still join `appnet` and pick apps from one list. Operators dock demanding class disks onto B/C and leave shared services on A.

Generalising to N

The three-Engine model is the **design target first**. N Engines later means:

- More class Engines (idea-D...), and/or
- More replicas of a hot app (extra Kolibri / Nextcloud instances) assigned by cohort.

Discovery, Console overview, and “users don’t pick an Engine” stay the same rules; only inventory and assignment grow.

Initial scenario list (for discussion)

Discussable outcomes — **not** UI click scripts. Each item maps to Markov **states** / **actions** (and thus a UI Interaction) where the usage graph covers it; items that are operator-side or not yet in the graph are marked explicitly.

Reason 1 — distribute apps / load

- **S1 Add Engine, plug demanding app:** School adds idea-C; docks a heavy Kolibri (or Nextcloud media) disk onto it; app appears in every Console's integrated list without renaming Consoles or teaching users a new hostname.
- **S2 Redistribute load:** Move or copy a demanding instance from idea-A onto idea-B so A stays responsive for shared services; app list and Open links remain coherent school-wide.
- **S3 Invisible Engine identity:** Student or teacher opens Console on idea-B or idea-C and sees the same app catalog/status as on idea-A.

Reason 2 — scale one app

- **S4 Kolibri stream ceiling:** Class exceeds safe concurrent streams on one instance → second Kolibri on another Engine; students assigned so neither instance exceeds the [idea#159](#) planning N for that video quality (64 @ 360p / 32 @ 480p–720p / 12 @ 1080p).
- **S5 Nextcloud media twin:** Same idea for Nextcloud video / large file concurrent access — second instance + cohort assignment when one Pi is the bottleneck.

Reason 3 — per-class Engine + Console-anywhere

- **S6 Class-only content:** Grade 5A Kolibri/Nextcloud live only on idea-B; Form 3 only on idea-C; shared Wikipedia (Kiwix) stays on idea-A.
- **S7 Manage from the other room:** Teacher or operator on idea-A's Console starts/stops, checks status, or prepares disks for apps that run on idea-B and idea-C.
- **S8 Cross-class share without moving rooms:** Teacher on idea-B Console shares a Nextcloud folder with Form 3's group whose home instance is on idea-C (management is school-wide; data placement follows class Engines).

Usage / field-aligned (Marco)

- **S9 Kolibri teaching cycle:** Teacher builds class → enrolls learners → assigns lesson with videos → students watch/complete → quiz → teacher reads Reports (from Marco Kolibri presentations).
- **S10 Nextcloud classroom workflow:** Class groups; view-only material share; File Drop homework; collaborative doc; Talk for support (from Marco Nextcloud presentations). Field note: many devices → performance drop → prefer one group at a time unless multi-Engine capacity is in play.

- **S11 Offline Wikipedia:** Learner (or teacher) opens Kiwix from Console (idea-A), searches/browses articles, dwells, leaves.

Coverage vs this usage graph (after rewrite): see [Initial scenarios ↔ Markov coverage](#).

Markov usage graph

Idea

- **States** = usage places (what a person is doing in an app or Console), not disk-dock hardware states alone.
- **Actions** = labeled transitions with probabilities (initial guesses until field frequencies exist). Action names are Intent-style explaining names (e.g. **Open Kolibri as teacher**, **Stay on teacher overview**, **Create class**) — no cryptic IDs and no `*` wildcards on the graph.
- A **real test** = a **random walk** on this graph for some duration or step count. Each action expands into a **UI Interaction** with the **same Intent name** (forward link only).
- **Wiring rule for prose:** arrival / entry UI (login, open app, land on home) belongs on the **outgoing action of the state you leave**, not as content of the destination. Destination state sections describe only what you do **while in** that state (outgoing actions from there).
- Test instances of each app ship with **preloaded content**; legal actions are derived from that content (e.g. which lessons/videos exist, which folders/groups exist).
- Incorporate Marco's file-sharing (students/teachers) and Kolibri classroom management, plus content-access actions for Kolibri, Nextcloud, and offline Wikipedia (Kiwix on idea-A).

Operator Console Markov (separate): managing/altering the multi-Engine setup (login, NetworkTree, dock/eject, install, start/stop, copy/move, Files/Backup/erase, operators, settings) lives in sibling [multi-engine-operator.md](#) with graph files `multi-engine-operator-markov.{dot,png,svg}`. This classroom usage graph stays usage-only.

States

State	Who	Colour	Meaning
<code>console_teacher</code>	Teacher	blue	School-wide Engine/app overview; opens apps as coach
<code>console_learner</code>	Student	mid-blue	Same unified app list; student opens an app to work

State	Who	Colour	Meaning
kolibri_home	Student or teacher	green	Kolibri signed in, Learn
kolibri_watching	Student	green	Playing an assigned lesson video
kolibri_exercise	Student	green	Doing an exercise in a lesson
kolibri_manage	Teacher	green (dark)	Classes, lessons, quizzes, Reports
nc_browse	Student or teacher	yellow	Nextcloud Files browse
nc_share	Teacher	yellow (dark)	View-only share to a class group
nc_drop	Student	yellow (dark)	Uploading to a File Drop
nc_collab	Student	yellow (dark)	Editing a shared collaborative doc
wiki_browse	Student or teacher	purple	Kiwix / offline Wikipedia on idea-A

Graph

The full state/action set is dense for one letter page, so the usage graph is shown as **two diagrams** (same states and actions; Console repeated as the shared hub). Combined source also kept: [multi-engine-markov.dot](#) → [multi-engine-markov.png](#) .

Every edge is labeled with the **same Intent / UI Interaction name** used in prose (plus ≈probability).

Diagram A — Console + Kolibri ([multi-engine-markov-kolibri.dot](#)):

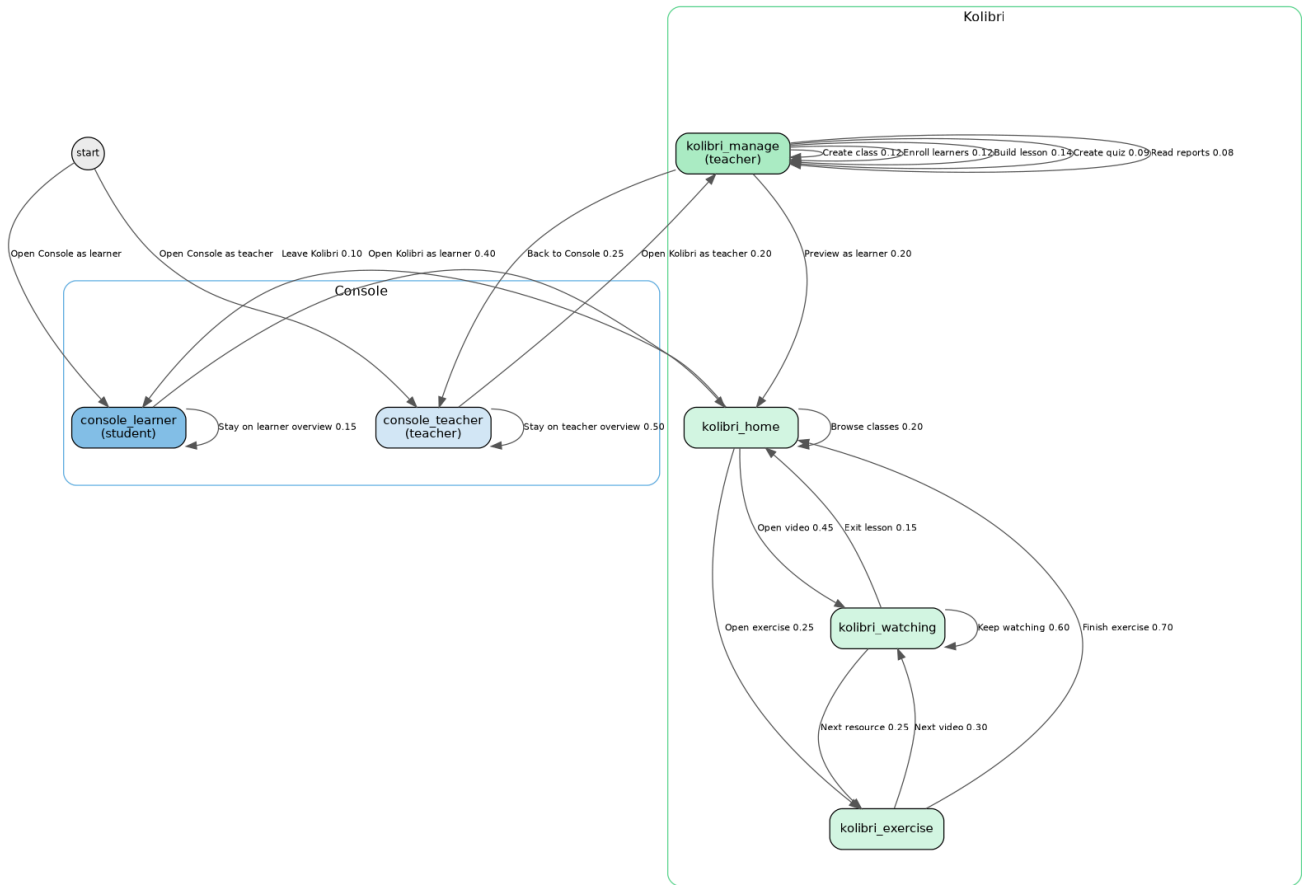
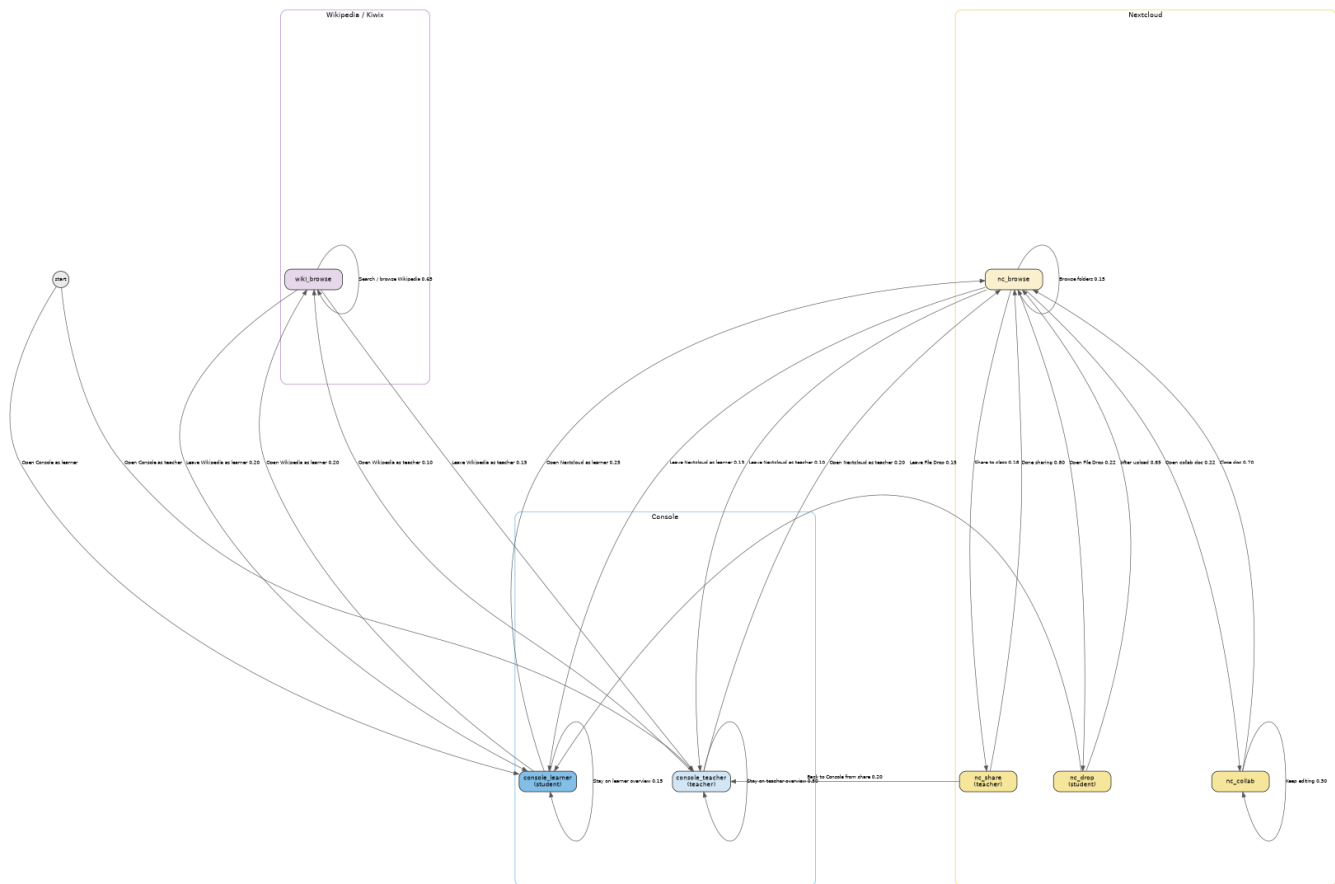


Diagram B — Console + Nextcloud + Wikipedia ([multi-engine-markov-files.dot](#)):



► Same graph as Mermaid (editable)

How states map to tests

Each Markov **state** is a coarse place someone can be. An **action** leaves that place for another (or stays via a self-loop). Inside a state, a real test does **not** invent a new state for every mouse click — that would explode the graph. Instead the harness expands the action into an **ordered UI Interaction** (login → sidebar → form → Save) taken from Marco's classroom presentations. Create an extra state only when the action changes **who** is acting, **which app surface** they are on, or **resource pressure** (e.g. video stream vs browse). Probabilities below are still placeholders.

Actions are unique to the state they depart from — an action from state A cannot leave or affect another state; you only take actions listed on the current state. Example: from `console_teacher` you may only **Open Kolibri as teacher**, **Open Nextcloud as teacher**, **Open Wikipedia as teacher**, or **Stay on teacher overview**. Coaching actions (**Create class**, **Enroll learners**, ...) exist only as actions inside `kolibri_manage`. Same pattern for Nextcloud/Wikipedia.

Arrival UI rule: login / open / land-on-home click sequences live on the **source state's outgoing action** (and its UI Interaction). Destination state sections never re-list how you got there.

Action names match Marco’s field decks and quick-reference cards (kolibri-classroom-setup , kolibri-lessons-and-quizzes , nextcloud-user-registration , nextcloud-classroom-use , quick-reference-kolibri.md , quick-reference-nextcloud.md) — confirmed with Marco 2026-09-29. Any Engine’s Console shows the same school-wide app list (reason 1 / 3). Detailed CSS selectors wait for the Playwright harness.

State: console_teacher

The teacher is on the **Engine / apps overview**: Engines present, apps docked, status. They are inspecting what is available on the network, not choosing a “home” Pi. **Operator-specific Console actions** (dock/eject disks, start/stop instances, install/move/backup/erase, operators/settings) are **not** on this usage graph — see multi-engine-operator.md .

While here: scan the integrated app list (Running, etc.) and Engine rows; optionally refresh / wait while students work.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open Kolibri as teacher	0.20	kolibri_manage	Open Kolibri as teacher
Open Nextcloud as teacher	0.20	nc_browse	Open Nextcloud as teacher
Open Wikipedia as teacher	0.10	wiki_browse	Open Wikipedia as teacher
Stay on teacher overview	0.50	console_teacher	Stay on teacher overview

State: console_learner

A student is on Console with the **same unified app list** as teachers (reason 1 / S3). They are not managing Engines — just picking Kolibri, Nextcloud, or Wikipedia.

While here: see integrated app list (class Kolibri/Nextcloud, Kiwix on idea-A, ...); choose an app or dwell.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open Kolibri as learner	0.40	<code>kolibri_home</code>	Open Kolibri as learner
Open Nextcloud as learner	0.25	<code>nc_browse</code>	Open Nextcloud as learner
Open Wikipedia as learner	0.20	<code>wiki_browse</code>	Open Wikipedia as learner
Stay on learner overview	0.15	<code>console_learner</code>	Stay on learner overview

State: `kolibri_manage`

The teacher is on Kolibri's **facility / coaching** side (Classes, Lessons, Quizzes, Reports) — not the learner Learn tab. Coaching Intents are **actions inside this state** (self-loops or leave).

While here: left sidebar visible; coach classes / lessons / quizzes / reports, or preview Learn / leave to Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
Create class	0.12	<code>kolibri_manage</code>	Create class
Enroll learners	0.12	<code>kolibri_manage</code>	Enroll learners
Build lesson	0.14	<code>kolibri_manage</code>	Build lesson
Create quiz	0.09	<code>kolibri_manage</code>	Create quiz
Read reports	0.08	<code>kolibri_manage</code>	Read reports
Preview as learner	0.20	<code>kolibri_home</code>	Preview as learner
Back to Console	0.25	<code>console_teacher</code>	Back to Console

Composite walks (Teacher prepares Grade 5A lesson, Quiz and reports) chain several of these — see [Composite walks](#).

State: `kolibri_home`

A student (or teacher previewing) is on the **learner** side after login — Learn tab, enrolled classes, assigned lessons.

While here: browse classes / lessons / quizzes; open a video or exercise; or leave Kolibri.

Actions from this state:

Action	≈p	Destination	UI Interaction
Open video	0.45	<code>kolibri_watching</code>	Open video
Open exercise	0.25	<code>kolibri_exercise</code>	Open exercise
Browse classes	0.20	<code>kolibri_home</code>	Browse classes
Leave Kolibri	0.10	<code>console_learner</code>	Leave Kolibri

State: `kolibri_watching`

Capacity-sensitive state (reason 2 / idea#159). Many concurrent walkers here is what forces a second Kolibri instance. Planning N depends on video quality: **64 @ 360p / 32 @ 480p–720p / 12 @ 1080p** (idea01 Pi 5 / 4 GB).

While here: video playing in-browser (no download); student may pause/seek; Kolibri records progress in the background.

Actions from this state:

Action	≈p	Destination	UI Interaction
Keep watching	0.60	<code>kolibri_watching</code>	Keep watching
Next resource	0.25	<code>kolibri_exercise</code>	Next resource
Exit lesson	0.15	<code>kolibri_home</code>	Exit lesson

State: `kolibri_exercise`

While here: exercise questions with immediate feedback; student answers and submits; Kolibri records started / completed / score.

Actions from this state:

Action	$\approx p$	Destination	UI Interaction
Finish exercise	0.70	<code>kolibri_home</code>	Finish exercise
Next video	0.30	<code>kolibri_watching</code>	Next video

(Quiz-taking can reuse exercise-like clicks under Learn → Quizzes, or stay under `kolibri_home` until a distinct load profile justifies a `kolibri_quiz` state. See [Composite walks](#).)

State: `nc_browse`

Hub for Marco's classroom file workflow. **Files** app is the central UI (manual p.12 in Marco's deck).

While here: browse folders (class materials, Drop Zone, school shares); start share / File Drop / collab; or leave to Console.

Actions from this state:

Action	$\approx p$	Destination	UI Interaction
Share to class	0.18	<code>nc_share</code>	Share to class
Open File Drop	0.22	<code>nc_drop</code>	Open File Drop
Open collab doc	0.22	<code>nc_collab</code>	Open collab doc
Browse folders	0.15	<code>nc_browse</code>	Browse folders
Leave Nextcloud as learner	0.13	<code>console_learner</code>	Leave Nextcloud as learner
Leave Nextcloud as teacher	0.10	<code>console_teacher</code>	Leave Nextcloud as teacher

State: `nc_share`

While here: Share dialog open on a folder; set internal group share to View only; finish or leave.

Actions from this state:

Action	≈p	Destination	UI Interaction
Done sharing	0.80	nc_browse	Done sharing
Back to Console from share	0.20	console_teacher	Back to Console from share

State: nc_drop

While here: File Drop / file-request page open; upload file(s); then return to Files or leave to Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
After upload	0.85	nc_browse	After upload
Leave File Drop	0.15	console_learner	Leave File Drop

State: nc_collab

While here: collaborative editor open on a shared doc; peer avatars / live cursors; edit or close.

Actions from this state:

Action	≈p	Destination	UI Interaction
Close doc	0.70	nc_browse	Close doc
Keep editing	0.30	nc_collab	Keep editing

State: wiki_browse

Shared service on **idea-A** (school golden hub). Light load relative to Kolibri video — still a first-class classroom usage state.

While here: Kiwix library / Wikipedia ZIM open; search or browse articles; leave to matching Console.

Actions from this state:

Action	≈p	Destination	UI Interaction
Search / browse Wikipedia	0.65	wiki_browse	Search / browse Wikipedia
Leave Wikipedia as learner	0.20	console_learner	Leave Wikipedia as learner
Leave Wikipedia as teacher	0.15	console_teacher	Leave Wikipedia as teacher

Initial scenarios ↔ Markov coverage

Initial item	In usage graph?	States / actions	UI Interaction(s)
S1 Add Engine, plug demanding app	Partial — usage asserts catalog equality; dock/install is operator	console_teacher / console_learner Stay on ... overview, Open Kolibri / Nextcloud / Wikipedia as ...	Stay on teacher overview , Stay on learner overview ; dock/install → operator graph
S2 Redistribute load	Not in usage graph yet — copy/move is operator Console	—	Operator: Copy app / Move app (multi-engine-operator.md)
S3 Invisible Engine identity	Yes	console_teacher , console_learner + open-app actions from any Engine's Console	Open Console as teacher , Open Console as learner , open-app UI Interactions
S4 Kolibri stream ceiling	Yes (usage dwell); second instance start is operator	kolibri_watching Keep watching (N walkers)	Keep watching
S5 Nextcloud media twin	Not in usage graph yet — no Nextcloud video-watching state; twin instance is operator	—	Aspirational until NC media dwell state is designed; operator install/start for second instance

Initial item	In usage graph?	States / actions	UI Interaction(s)
S6 Class-only content	Yes (placement story; same states, class Engines)	Open Kolibri/Nextcloud/Wikipedia from Console toward class vs shared instances	Open-app UI Interactions; Search / browse Wikipedia
S7 Manage from the other room	Partial — teacher open-app from idea-A Console yes ; operator start/stop/disk operator graph	<code>console_teacher</code> Open Nextcloud as teacher / Open Kolibri as teacher while app runs on B/C	Open Nextcloud as teacher , Open Kolibri as teacher ; operator start/stop elsewhere
S8 Cross-class share	Yes	<code>nc_browse</code> Share to class → <code>nc_share</code>	Share to class , Done sharing
S9 Kolibri teaching cycle	Yes	<code>kolibri_manage</code> coaching actions + learner path	Create class ... Read reports , Open video , Open exercise ; Composite walks
S10 Nextcloud classroom workflow	Yes (Talk optional / light)	<code>nc_browse</code> , <code>nc_share</code> , <code>nc_drop</code> , <code>nc_collab</code>	Share to class , Open File Drop , Open collab doc ; Composite walks
S11 Offline Wikipedia	Yes	<code>wiki_browse</code> Search / browse Wikipedia	Search / browse Wikipedia

UI Interactions

These are **ordered UI click sequences** that implement one graph **action**. Each has a single consistent Intent-style name (no cryptic IDs). Markov actions link **forward** to these names only; this chapter does **not** re-describe the Markov graph.

Preload content so every click target exists.

Open Console as teacher

Role: teacher · **Engines:** any Console (idea-A / B / C equivalent).

1. Open Chromium (or any browser) on the school LAN → Console for any Engine (field habit today: hostname like `engine-1.local` ; multi-Engine story: idea-A / idea-B / idea-C Consoles must be equivalent).
2. Land on Engine / apps overview (Running apps, Engine rows).

Open Console as learner

Role: student · **Engines:** any Console.

1. Open browser on school LAN → any Engine's Console.
2. Land on unified app list (same catalog as teachers).

Stay on teacher overview

Role: teacher · Dwell on school-wide Console overview; refresh while students work; assert catalog visible without opening an app.

Stay on learner overview

Role: student · Dwell on unified app list without opening an app yet.

Open Kolibri as teacher

Role: teacher · Arrives in `kolibri_manage` .

1. From Console overview → click Kolibri (Running).
2. Kolibri login → type **teacher username** / **password** → Sign in.
3. Land on teacher home / facility UI (left sidebar visible) — ready for coaching actions.

Open Kolibri as learner

Role: student · Arrives in `kolibri_home` .

1. From Console overview → click Kolibri (Running).
2. Kolibri login → **learner username** / **password** → Sign in.
3. Top menu **Learn** → see enrolled classes / assigned Lessons (and Quizzes if active).

Open Nextcloud as teacher

Role: teacher · Arrives in `nc_browse` .

1. From Console overview → click Nextcloud (Running).

2. Nextcloud login → teacher username / password.
3. Open **Files** (default home) → ready to browse / share.

Open Nextcloud as learner

Role: student · Arrives in `nc_browse` .

1. From Console overview → click Nextcloud (Running).
2. Nextcloud login → student username / password.
3. Open **Files** → class materials / Drop Zone / shared docs.

Open Wikipedia as teacher

Role: teacher · Arrives in `wiki_browse` .

1. From Console overview → click **Kiwix** / Wikipedia app (Running on idea-A).
2. Kiwix library / Wikipedia ZIM opens in the browser.

Open Wikipedia as learner

Role: student · Arrives in `wiki_browse` .

1. From Console overview → click **Kiwix** / Wikipedia app (Running on idea-A).
2. Kiwix library / Wikipedia ZIM opens in the browser.

Create class

Role: teacher · *Requires already in* `kolibri_manage` .

1. Left sidebar **Classes** → **+ New class** → type name (e.g. `Grade 5A`) → **Save**.

Enroll learners

Role: teacher · *Requires class (preload or Create class).*

1. Open class → **Learners** tab → **Enroll learners** → tick students → **Confirm**.

Build lesson

Role: teacher

1. Class → **Lessons** → **+ New lesson** (or open existing) → name (e.g. `Week 1: Fractions Introduction`).

2. **Add resources** → Library/Channels → select **1 video + 1 exercise** → **Save**.
3. Confirm **Recipients** = class → toggle lesson **Visible**.

Create quiz

Role: teacher

1. Class → **Quizzes** → **+ New quiz** → **Add questions** from exercise channel → set count (e.g. 5–10) → **Finish** → toggle **Active**.

Read reports

Role: teacher

1. Left sidebar **Reports** → **Classes** → class → **Lessons** or **Quizzes** → open item → scan learner table (scores / completion).

Preview as learner

Role: teacher · Leaves coaching for Learn (`kolibri_home`).

1. From coaching UI, open learner view / Learn (or sign in as a test learner).
2. See assigned lessons as a student would.

Back to Console

Role: teacher · Leaves Kolibri coaching for `console_teacher` .

1. Close / navigate away from Kolibri → Console overview visible again.

Open video

Role: student · From Learn / lesson list.

1. Open class → open assigned lesson.
2. Click video resource → player starts (enters `kolibri_watching`).

Open exercise

Role: student

1. Open class → open assigned lesson (or from lesson list).
2. Click exercise resource → exercise UI loads (enters `kolibri_exercise`).

Browse classes

Role: student · Stay in `kolibri_home` .

1. On Learn, open / close classes, scan assigned Lessons and Quizzes without starting a resource.

Leave Kolibri

Role: student · Returns to `console_learner` .

1. Leave Kolibri (close tab / navigate back) → Console unified app list.

Keep watching

Role: student · Self-loop in `kolibri_watching` .

1. Continue playing / pause / seek; Kolibri records progress.
2. *Concurrency note (S4):* N walkers dwelling here = N streams; if N > planning N for that quality on idea-B, second Kolibri on idea-C is an **operator** action.

Next resource

Role: student · Video → exercise in the same lesson.

1. Finish or leave video player → open the exercise resource in the lesson → `kolibri_exercise` .

Exit lesson

Role: student · Returns to `kolibri_home` .

1. Exit lesson / back to Learn or class list.

Finish exercise

Role: student · Returns to `kolibri_home` .

1. Answer remaining items → submit → return to Learn / class.

Next video

Role: student · Exercise → another video.

1. From exercise (or lesson list) open the next video resource → `kolibri_watching` .

Share to class

Role: teacher · Enters `nc_share` .

1. In Files, hover the folder (e.g. `Videos to watch`).
2. Click **Share** (person + icon).
3. Under **Internal shares**, type class group name (e.g. `Grade 5A`) → select group.
4. Set permission **View only**.

(Group must already exist — Accounts → Groups → + — often a preload step.)

Done sharing

Role: teacher · Returns to `nc_browse` .

1. Close Share dialog → back in Files browse.

Back to Console from share

Role: teacher · Leaves Nextcloud for `console_teacher` .

1. From Share / Files, leave Nextcloud → Console overview.

Open File Drop

Role: student (teacher prep optional) · Enters `nc_drop` .

1. *(Teacher prep / preload, often outside the walk:)* `Drop Zone` → Share → public link + → **File request** → copy link.
2. Student: open the File Drop / file-request link.
3. See empty “Click or drop to upload” UI (cannot see others’ files).

After upload

Role: student · Returns to `nc_browse` .

1. Choose file(s) / drop → upload completes → return to Files browse.

Leave File Drop

Role: student · Returns to `console_learner` .

1. Leave File Drop page → Console.

Open collab doc

Role: student (teacher create often preload) · Enters `nc_collab` .

1. (*Teacher create / preload:*) Files → **+ New** → **New Document** → name → pick template → Share with group → **Allow editing**.
2. Student: open shared file from Files → editor loads; peer **avatars** top-right.

Close doc

Role: student · Returns to `nc_browse` .

1. Close editor → Files browse.

Keep editing

Role: student · Self-loop in `nc_collab` .

1. Continue typing / collaborating; cursors move live.

Browse folders

Role: student or teacher · Self-loop in `nc_browse` .

1. Navigate folders (class materials, Drop Zone, school shares) without starting share / drop / collab.

Leave Nextcloud as learner

Role: student · Returns to `console_learner` .

1. Leave Nextcloud → Console unified app list.

Leave Nextcloud as teacher

Role: teacher · Returns to `console_teacher` .

1. Leave Nextcloud → Console overview.

Search / browse Wikipedia

Role: learner or teacher · Self-loop in `wiki_browse` .

1. Search box: type a topic → open an article.
2. Follow internal links / browse categories; dwell reading.

Leave Wikipedia as learner

Role: student · Returns to `console_learner` .

- 1. Leave Kiwix → Console.

Leave Wikipedia as teacher

Role: teacher · Returns to `console_teacher` .

- 1. Leave Kiwix → Console.

Composite walks

Optional multi-action stories for discussion (not extra graph edges). Each step is a named UI Interaction above.

Walk	Chain
Teacher prepares Grade 5A lesson	Open Kolibri as teacher → Create class (<i>skip if preloaded</i>) → Enroll learners → Build lesson → optional Read reports → Preview as learner or Back to Console
Student completes assigned lesson	Open Console as learner → Open Kolibri as learner → Open video → Keep watching → Next resource → Finish exercise → Leave Kolibri
Quiz and reports	Create quiz → (<i>students: Learn → Quizzes → Start → answer</i>) → Read reports
Teacher share File Drop and collab	Open Nextcloud as teacher → Share to class → (<i>prep File Drop / create collab, often preload</i>) → students: Open File Drop / Open collab doc
Class hits stream ceiling	N × (Open Kolibri as learner → Open video → Keep watching); second instance start is operator

How a walk becomes a test

- 1. Preload Kolibri with a class, enrolled learners, and a lesson that includes at least one video and one exercise (Marco Week 1–2 content shape).
- 2. Preload Nextcloud with class groups, a view-only materials folder, a File Drop folder, and one collaborative document (Marco classroom-use shape).
- 3. Preload / confirm Kiwix ZIM available on idea-A.
- 4. Sample a walk: e.g. `console_learner` → `kolibri_home` → ... → `kolibri_watching` (N concurrent walkers = N “students” hitting video — bounded by idea#159 quality table), or `console_teacher` → `kolibri_manage` for coaching scripts.

5. Multi-Engine walks assign walkers to instances on idea-B vs idea-C when testing reason 2; Console actions may start on any Engine's Console (reason 3).

Full YAML / runner design stays with the existing Markov duration-tests proposal ([duration-tests.md](#) , idea#85) when implementation starts — not here.

Sources consulted

Source	Use
agent-engine-dev/proposals/solution-description.md	Vision: autofind Appdockers; add Appdockers + redistribute apps for performance
multi-engine-classroom-scenarios.md (+ .pdf)	Prior draft — superseded (implementation-heavy; incorrectly framed multi-Engine as undocumented)
Marco / programme-manager: presentations/kolibri-classroom-setup , kolibri-lessons-and-quizzes , nextcloud-user-registration , nextcloud-classroom-use ; reference-cards/quick-reference-kolibri.md , quick-reference-nextcloud.md	Exact action names / UI wording for Markov story + UI Interactions (Classes, Learn, Share, File request, ...)
Marco field: field/troubleshooting-guide.md (via prior draft + local copy)	Many devices → performance drop; limit one group at a time
duration-tests.md	Existing Markov duration-test design (implementation later)
idea#159 (+ multi-quality comment)	Kolibri concurrent stream capacity — planning 64/32/32/12 by quality

Out of scope for now

Flagged for later proposals / issues — **not** designed here:

- Implementation of multi-Engine discovery / peering changes

- Auth, operator roles, and student-assignment UX between Kolibri instances
- Playwright / Console e2e harness, fleet claim scripts, fixture disks
- Exact Engine APIs for “assign students across instances”
- Hardware power-cut, USB disk carry automation, golden-Pi rules
- PDF/print packaging of field guides
- **~~Operator Console Markov graph~~ — done as sibling:** `multi-engine-operator.md` + `multi-engine-operator-markov.*` (kept separate from this usage graph)

When Koen agrees the shape, Steve can split follow-ups (Engine / Console / App Dev / Ops) without baking premature design into this doc.

Ask of Koen

1. Confirm the **three reasons** match intent.
2. Confirm **idea-A / idea-B / idea-C** as the named 3-Engine story (or rename).
3. Confirm `console_teacher` + `console_learner` + `wiki_browse` and coaching actions only inside `kolibri_manage`.
4. Confirm **UI Interactions** naming (action name = edge label = chapter heading) and arrival UI only on source actions.
5. Prioritise which of S1–S11 to deepen next; review the separate Operator Console Markov (`multi-engine-operator.md`).
6. Treat idea#159 quality table (64/32/32/12) as the capacity gate for reason-2 scenarios.